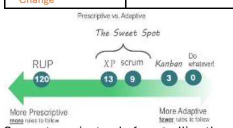
Agile Software Project Management: A project manager is responsible for **planning**, **scheduling**, **coordinating** project activities, and **working with people** to ensure that the project objectives are achieved successfully. A project phase is defined by the completion and approval of one or more deliverables. In iterative development, a new phase may begin before the previous phase is completely closed. The initiation phase is the starting phase of a project where the project idea, objectives, business need, and major stakeholders are identified. The project is formally approved during this phase. The planning phase defines how the project will be completed by preparing plans for scope, schedule, cost, resources, risk, quality, and communication. The execution phase is where the project team performs the planned work and produces the project deliverables. The monitoring and controlling phase tracks project progress, compares actual performance with the plan, manages changes, and takes corrective actions when needed. The closing phase finalizes all project activities, obtains customer acceptance, documents lessons learned, releases resources, and formally closes the project. Organizational Structures. Organizational Structures:

Structure	Form / How it works	Pros	Cons
Functional Organization	Employees are grouped by department/function. Example: IT, HR, Finance, Marketing. Project work is done inside functional departments. Functional manager has more authority.	Clear department structure. Strong specialization. Easy resource supervision inside departments.	Project manager has less authority. Project work can be slow. Communication between departments can be difficult.
Projectized Organization	Team members are organized around projects. Project manager has high authority. Team members mainly report to the project manager during the project.	Fast decision-making. Strong project focus. Clear project responsibility. Better team communication.	Resources may be duplicated across projects. Team members may not have a clear department after project ends. Can be costly.
Matrix Organization	Combination of Functional and Projectized structures. Team members report to both functional manager and project manager.	Better resource sharing. Functional expertise + project focus both available. Flexible for multiple projects.	Dual reporting can create confusion. Conflicts can happen between functional manager and project manager. Decision-making can be slower.

Agile transformation is the process of moving from traditional project management practices to Agile ways of working, where teams focus on flexibility, collaboration, customer feedback, and incremental delivery.

Agile Manifesto Value	Preference Shown in Lecture Slide	Simple Explanation
Individuals & Interactions	Over Tools & Processes	Agile values teamwork, communication, and collaboration more than only depending on tools or fixed processes.
Working Software	Over Comprehensive Documentation	Agile focuses more on delivering usable working software rather than spending too much time on large documentation.
Customer Collaboration	Over Contract Negotiation	Agile encourages regular communication and feedback from the customer instead of only following fixed contract terms.
Responding to Change	Over Following a Plan	Agile supports flexibility. If requirements change, the team responds to the change instead of strictly following the original plan.



The Scrum Master is a servant leader who supports the Scrum team, removes blockers, facilitates Scrum events, and protects the project by ensuring that Scrum principles are followed.

Servant Leader: A Scrum Master acts as a servant leader by supporting the Scrum team instead of controlling them. They help the team work smoothly, follow Scrum practices, and improve collaboration. Resolves Blockers: The Scrum Master helps remove problems that stop the team from completing their work. These blockers can be technical issues, unclear requirements, lack of resources, or communication problems. Guardian of the Project: The Scrum Master protects the project by making sure the team follows Agile and Scrum principles. They also protect the team from unnecessary interruptions and help keep the sprint goal on track. Scrum Framework:

Scrum Area	Simple Meaning	Key Details
Product Owner	The person responsible for the product vision and requirements.	The Product Owner manages the product backlog, prioritizes user stories, represents customer/business needs, and decides what should be built first.
Scrum Master	The person who supports the Scrum team and makes sure Scrum is followed properly.	The Scrum Master acts as a servant leader, removes blockers, facilitates Scrum events, and protects the team from unnecessary interruptions.
Development Team	The team that builds the actual product.	The Development Team completes the selected sprint work and delivers a working product increment at the end of the sprint.
Product Backlog	A prioritized list of everything needed for the product.	It can include features, requirements, improvements, bug fixes, and technical tasks. The Product Owner mainly manages and prioritizes it.
Sprint	A fixed time period where the team completes selected work.	A sprint is usually 1–4 weeks. The team works on selected backlog items and delivers a working increment.
Sprint Planning	A meeting held at the start of the sprint.	The team selects product backlog items for the sprint, defines the sprint goal, and creates the sprint backlog.
Sprint Backlog	The list of work selected for the current sprint.	It includes the product backlog items that the team commits to complete during that sprint.
Increment	The working output delivered at the end of a sprint.	It should be usable and potentially releasable. Each sprint should add value to the product.

A user story describes a requirement from the user's perspective using the format: As a persona, I want something, so that I can achieve a benefit. Backlog grooming is the process of reviewing and refining the product backlog by removing irrelevant stories, adding new stories, re-prioritizing items, estimating stories, correcting estimates, and splitting large stories into smaller manageable stories. Planning Poker is a gamified Agile estimation technique where the Scrum team estimates product backlog items using cards or a mobile application, usually based on relative estimation. Planning Poker Steps: 1) The Scrum team gathers with planning poker cards or a planning poker mobile application. 2) The Scrum Master selects a product backlog item and discusses it with the team so everyone understands the requirement clearly. 3) Each team member privately selects an estimate based on the effort, size, or complexity of the backlog item. 4) All team members reveal their estimates at the same time when the Scrum Master gives the signal. 5) If the estimates are different**,** the team members with the highest and lowest estimates explain the reasons behind their values. 6) The team discusses the differences to understand risks, complexity, unclear requirements, or hidden work. 7) The team estimates again after the discussion. 8) The process repeats until the estimates become closer or the majority of the team agrees on one estimate. 9) The team moves to the next backlog item and repeats the same process. 10) Planning Poker continues until enough backlog items are estimated for the sprint. Daily Scrum is a short daily meeting where team members discuss what they did yesterday, what they will do today, and any impediments blocking their progress.

Project Selection and Integration Management: Project Selection – how an organization chooses the best project. Project Integration Management – how the project manager coordinates all parts of the project together. Strategic planning means deciding the long-term objectives of an organization, predicting future trends, and identifying the need for new products or services. As part of strategic planning, organizations identify potential projects, use realistic methods to select the best projects, and formally start a selected project by issuing a project charter. Methods for selecting projects include: • focusing on broad organizational needs • categorizing information technology projects • performing net present value or other financial analyses • using a weighted scoring model • implementing a balanced scorecard. SWOT Analysis

SWOT Area	Meaning
Strengths	Internal advantages of the organization
Weaknesses	Internal limitations or problems
Opportunities	External chances that the organization can benefit from
Threats	External risks or challenges that can negatively affect the organization

A project can be selected based on broad organizational needs if there is a clear need for the project, funds are available, and there is strong organizational support to make the project successful.

Categorizing IT Projects: **Problem** A project that solves an existing issue. Example: A company's current system is slow, so they start a project to build a faster system. **Opportunity** A project that helps the organization gain a new benefit. Example: A travel company builds a mobile app to reach more customers. **Directive** A project that is required because of a rule, law, regulation, or management instruction. Example: A company updates its system to follow new data protection rules. Financial Analysis of Projects: NPV is used to calculate the expected net monetary gain or loss from a project by converting future cash inflows and outflows into present value. If financial value is a key criterion, projects with positive NPV should be considered, and the higher the NPV, the better. Positive NPV = project should be considered if financial value is important. Higher NPV = better project. ROI measures the return earned from a project compared to its cost. The higher the ROI, the better the project is financially. ROI = (Total discounted benefits - Total discounted costs) / Discounted costs. Payback analysis measures how long it takes to recover the total investment of a project through net cash inflows. Organizations usually prefer projects with shorter payback periods. A Weighted Scoring Model is a tool that provides a systematic process for selecting projects based on many criteria. Weighted Scoring Model Steps: 1)Identify the criteria important for project selection. 2)Assign weights to each criterion as percentages. 3)Make sure all weights add up to 100%. 4)Assign scores to each project for each criterion. 5)Multiply each score by its weight. 6)Add the weighted scores to get the total weighted score. 7)The project with the higher weighted score is better. A balanced scorecard helps organizations select and manage projects that align with business strategy by converting value drivers such as customer service, innovation, operational efficiency, and financial performance into measurable metrics. Project Integration Management involves coordinating all project management knowledge areas throughout the project life cycle to ensure that the project works as one complete and successful effort. Project Integration Management Processes. Develop Project Charter Create the document that formally authorizes the project and gives the project manager authority to start the project. Develop Project Management Plan Prepare one complete project management plan by combining all planning areas such as scope, schedule, cost, quality, risk, communication, and resources. Direct and Manage Project Work Carry out the project work according to the project management plan and produce the project deliverables. Monitor and Control Project Work Track the project progress, compare actual performance with the plan, and take corrective actions when needed. Perform Integrated Change Control Review, approve, reject, and manage changes that can affect the project scope, time, cost, quality, or risks. Close Project or Phase Finalize all project activities, obtain acceptance for deliverables, complete documentation, and formally close the project or phase. Performing Integrated Change Control Integrated change control has three main objectives: 1)Influencing the factors that create changes to ensure changes are beneficial. 2)Determining that a change has occurred. 3)Managing actual changes as they occur. Change Control on IT Projects: Former view The project team should do exactly what was planned, on time and within budget. Problem with former view Stakeholders rarely agree fully on project scope at the beginning, and time and cost estimates can be inaccurate. Modern view Project management is a process of constant communication and negotiation. Solution Changes are often beneficial, and the project team should plan for them. A change control system is a formal documented process that explains when and how project documents and work can be changed, who can approve changes, and how changes should be made. A Change Control Board is a formal group responsible for reviewing, approving, or rejecting project change requests and managing the implementation of approved changes. A **48-hour policy**, where project team members can make decisions, and senior management has 48 hours to reverse the decision. Configuration management ensures that the descriptions of the project's products are correct and complete. It involves identifying and controlling the functional and physical design characteristics of products and support documentation.

Formula Name / Method	Formula
Return on Investment (ROI)	**ROI = (Total Discounted Benefits – Total Discounted Costs) / Discounted Costs**
Net Present Value (NPV)	**NPV = Total Present Value of Benefits – Total Present Value of Costs**
Internal Rate of Return (IRR)	**IRR = Discount Rate where NPV = 0**
Payback Period	**Payback Period = Time taken to recover the initial investment**
Weighted Score	**Weighted Score = Score × Weight**
Total Weighted Score	**Total Weighted Score = Sum of all Weighted Scores**

Project Cost Management: Cost is a resource sacrificed or given up to achieve a specific objective. In projects, costs are usually measured in money, such as rupees or dollars. Project Cost Management helps the project manager plan, estimate, budget, and control project costs so that the project can be completed within the approved budget. Project Cost Management Processes: Plan Cost Management Deciding the policies, procedures, and documentation that will be used for planning, executing, and controlling project costs. Estimate Costs Developing an approximate estimate of the resources and costs needed to complete the project. Determine Budget Allocating the overall cost estimate to individual work items and creating a cost baseline. Control Costs Monitoring cost performance and controlling changes to the project budget. Basic Principles of Cost Management Project managers should understand financial terms because executives usually understand financial language more than technical language. The lecture mentions these important principles: **Profit = Revenues – Expenditures,** Profit means the money remaining after subtracting expenses from revenue. **Profit margin** is the ratio of revenues to profits. **Life cycle costing** considers the total cost of ownership, including development cost and support/maintenance cost. **Cash flow analysis** determines estimated annual costs and benefits and the resulting annual cash flow for a project. Types of Costs and Benefits**:** Tangible Costs or Benefits These are costs or benefits that can be easily measured in monetary terms. Example: hardware cost, software license cost, salary cost, sales revenue. Intangible Costs or Benefits These are difficult to measure in monetary terms. Example: improved customer satisfaction, better company image, improved employee morale. Direct Costs These are costs directly related to producing the project product or service. Example: developer salary for the project, project-specific software tools. Indirect Costs These are costs not directly related to one project, but indirectly related to performing the project. Example: electricity, office rent, internet, admin support. Sunk Cost Sunk cost is money already spent in the past. When deciding whether to continue or invest in a project, sunk costs should not be included. **Reserves** are amounts included in a cost estimate to handle cost risk and future uncertain situations. Contingency Reserves are for future situations that may be partially planned for. These are also called known unknowns and are included in the project cost baseline. Example: possible price increase of software licenses. Management Reserves are for unpredictable future situations. These are called unknown unknowns. Example: sudden legal requirement or unexpected major economic issue. Planning Cost Management**:** The project team uses expert judgment, analytical techniques, and meetings to develop the cost management plan. A cost management plan includes: Level of accuracy and units of measure | Organizational procedure links | Control thresholds | Rules of performance measurement | Reporting formats | Process descriptions

Estimate Type	Accuracy Range	When Used
Rough Order of Magnitude — ROM	**-50% to +100%**	Early stage of the project
Budgetary Estimate	**-10% to +25%**	During budgeting/planning
Definitive Estimate	**-5% to +10%**	Later stage when details are clearer

Types of Cost Estimates

Cost Estimation Tools and Techniques: Analogous / Top-down Estimate This uses the actual cost of a previous similar project as the basis for estimating the current project cost. Example: If a similar mobile app cost Rs. 1 million, the new app may be estimated around that amount. Bottom-up Estimate This estimates individual work items or activities and then adds them together to get the total project cost. Example: UI design + backend + database + testing + deployment = total cost. Parametric Modeling This uses project characteristics or parameters in a mathematical model to estimate cost. Example: Cost per screen × number of screens. Main EVM Terms : Planned Value PV is the authorized budget assigned to scheduled work. Simple meaning: **How much work should have been completed according to the plan?** Earned Value EV is the value of work actually completed, expressed in terms of the budget authorized for that work. Simple meaning: **How much budgeted value has actually been earned by completed work?** Actual Cost AC is the actual cost spent for the work performed during a specific time period. Simple meaning: **How much money has actually been spent?** Budget at Completion BAC is the total approved budget for the project.

Formula Name	Formula
Earned Value — EV	**EV = % Completed × BAC**
Schedule Variance — SV	**SV = EV - PV**
Cost Variance — CV	**CV = EV - AC**
Variance at Completion — VAC	**VAC = BAC - AC**
Schedule Performance Index — SPI	**SPI = EV / PV**
Cost Performance Index — CPI	**CPI = EV / AC**
Estimate at Completion — EAC 1	**EAC = AC + ETC**
EAC 2: ETC at budgeted rate	**EAC = AC + (BAC - EV)**
EAC 3: ETC at present CPI	**EAC = BAC / CPI**
EAC 4: considering SPI and CPI	**EAC = AC + [(BAC - EV) / (CPI × SPI)]**
TCPI	**TCPI = Work Remaining / Funds Remaining**
Estimated Duration to Complete	**Estimated Duration = Original Duration / SPI**

CV Value	Meaning
Positive CV	**Under budget / cost saving**
Negative CV	**Over budget**
Zero CV	**On budget**

SV Value	Meaning
Positive SV	**Ahead of schedule**
Negative SV	**Behind schedule**
Zero SV	**On schedule**

CPI Value	Meaning
CPI > 1	**Good cost performance / under budget**
CPI < 1	**Poor cost performance / over budget**
CPI = 1	**On budget**

SPI Value	Meaning
SPI > 1	**Ahead of schedule**
SPI < 1	**Behind schedule**
SPI = 1	**On schedule**

Question Wording	Use Formula
ETC is given	**EAC = AC + ETC**
Variance is expected to stop / future work at budgeted rate	**EAC = AC + (BAC - EV)**
Current CPI will continue / cost variance continues	**EAC = BAC / CPI**
Both CPI and SPI affect remaining work	**EAC = AC + [(BAC - EV)/(CPI × SPI)]**

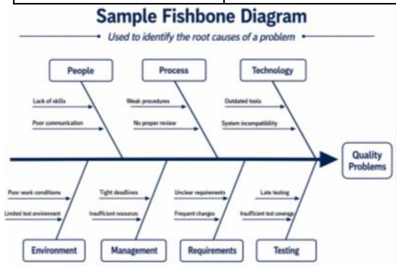
TCPI Value	Meaning
TCPI = 1.0	**Right on budget**
TCPI > 1.0	**Need stricter cost management or budget may be exceeded**
TCPI < 1.0	**Within budget and performing well**

Project Quality Management: Quality means the degree to which a product or project meets requirements. According to the lecture, quality can also be understood through two ideas: conformance to requirements and fitness for use. **Conformance** to requirements means the project processes and products meet the written specifications. **Fitness for use** means the product can be used for the purpose it was intended for. Project Quality Management Processes**:** Planning Quality Management This means identifying which quality standards are relevant to the project and deciding how to satisfy them. A metric is a standard of measurement. Performing Quality Assurance This means periodically evaluating overall project performance to ensure the project will satisfy relevant quality standards. Performing Quality Control This means monitoring specific project results to ensure they comply with the relevant quality standards. Planning quality means anticipating situations and preparing actions to achieve the desired quality outcome. Quality Assurance vs Quality Control

Area	Simple Meaning	Focus
Quality Assurance	**Evaluating overall project performance to make sure quality standards will be met.**	**Process-focused**
Quality Control	**Checking specific project results to confirm they meet standards.**	**Product/result-focused**

Scope Aspects of IT Projects The lecture connects quality with IT project scope. These are the key scope aspects: Functionality The degree to which a system performs its intended function. Features The special characteristics of the system that appeal to users. System Outputs The screens and reports generated by the system. Performance How well the product or service performs the customer's intended use. Reliability The ability of the product or service to perform as expected under normal conditions. Maintainability The ease of performing maintenance on the product. Controlling Quality The main outputs of quality control are: **Acceptance Decisions** | **Rework** |**Process Adjustments**

Seven Basic Tools of Quality	Simple Meaning
Cause-and-Effect Diagram / Fishbone	Used to identify the root causes of a quality problem.
Control Chart	Used to check whether a process is in control or out of control.
Checksheet	Used to collect and record data in an organized way.
Scatter Diagram	Used to show the relationship between two variables.
Histogram	Used to show frequency distribution using bars.
Pareto Chart	Used to identify and prioritize the most important problem areas.
Flowchart	Used to show the steps and flow of a process.



Statistical Sampling means choosing part of a population for inspection. The sample size depends on how representative the sample should be. Testing should not be done only at the end. It should be performed throughout the IT product development life cycle to detect issues early. ISO 9000 is a quality system standard. It is a three-part continuous cycle of planning, controlling, and documenting quality in an organization. It provides minimum requirements for quality certification and helps organizations reduce costs and improve customer satisfaction

Improving IT Project Quality: Establish leadership that promotes quality. | Understand the cost of quality. | Focus on organizational influences and workplace factors that affect quality. | Follow maturity models. Five Cost Categories Related to Quality**:** Prevention Cost Cost of planning and executing the project so it is error-free or within an acceptable error range. Appraisal Cost Cost of evaluating processes and outputs to ensure quality. Internal Failure Cost incurred to correct a defect before the customer receives the product. External Failure Cost Cost related to errors not detected and corrected before delivery to the customer. Measurement and Test Equipment Costs Capital cost of equipment used for prevention and appraisal activities.

Project Communication Management: Importance of Good Communication Good communication is very important in IT projects because the lecture says the greatest threat to many projects is a failure to communicate. IT professionals also need strong verbal and non-technical communication skills to succeed in their careers. Project Communications Management Processes Project Communication Management has three main processes: Planning Communications Management Determining the information and communication needs of stakeholders. Managing Communications Creating, distributing, storing, retrieving, and disposing of project communications according to the communication management plan. Controlling Communications Monitoring and controlling project communications to ensure stakeholder communication needs are met. Importance of Face-to-Face Communication: Body Language 58% + Tone 35% + Words 7%. Communication in Agile and Hybrid Environments**:** **Daily stand-ups** support frequent and open communication. **| Information radiators** and **feedback loops** support transparency and adaptation. Communication strategies should be tailored to Agile, Hybrid, and Predictive contexts Communication can also become more complex because of: Different working hours | Language barriers | Different cultural norms | Geographic location differences . **Number of Communication Channels = $n(n - 1) / 2$,** Where n = number of people involved. A communications management plan explains how project information will be created, shared, stored, updated, and communicated to stakeholders. Communications Management Plan Contents: Stakeholder communication requirements | Information to be communicated, including format, content, and level of detail | Who will receive the information and who will produce it | Suggested methods or technologies for conveying the information | Frequency of communication | Escalation procedures for resolving issues | Revision procedures for updating the communications management plan | Glossary of common terminology. Improving Team Collaboration and Communication: Promote transparency and accountability. | Set clear expectations and goals. | Establish effective communication channels and protocols. | Provide regular feedback and recognition. | Foster trust, respect, and mutual support. | Facilitate team-building activities and social interaction. Managing communications ensures that the right information reaches the right stakeholders at the right time and in the right format.

Report Type	Meaning
Status Report	Describes where the project stands at a specific point in time.
Progress Report	Describes what the project team has accomplished during a certain period.
Forecast	Predicts future project status and progress based on past information and trends.

Project Human Resource Management: Project Human Resource Management is important because project success depends heavily on the people involved, their skills, teamwork, motivation, and performance. Project Human Resource Management helps the project manager plan, obtain, develop, and manage the people and resources needed to complete the project successfully. Project Human Resource Management Processes**:** Plan Human Resource Management Identifying and documenting project roles, responsibilities, and reporting relationships. Acquire Project Team Getting the required people assigned to and working on the project. Develop Project Team Building individual and group skills to improve project performance. Manage Project Team Tracking team member performance, motivating team members, providing feedback, resolving issues and conflicts, and coordinating changes to improve project performance. The People Performance Domain focuses on creating a strong team culture through leadership, collaboration, trust, empowerment, and stakeholder engagement. Keys to Managing People: Motivation theories | Influence and power | Effectiveness. Developing the human resource plan means identifying and documenting project roles, responsibilities, and reporting relationships. The HR plan contents include: Project organizational charts | Staffing management plan | Responsibility assignment matrices | Resource histograms. A project organizational chart shows the reporting relationships of project team members. A staffing management plan describes how and when people will be added to and removed from the project team. A Responsibility Assignment Matrix (RAM) maps the work of the project from the WBS to the people responsible for doing the work from the OBS. A RAM shows which person or role is responsible for each project task or work package.

Letter	Meaning
R	Responsibility
A	Accountability — only one A per task
C	Consultation
I	Informed

Tools and Techniques for Planning Human Resource Management
The lecture lists these tools and techniques: Organization charts and position descriptions | Networking | Organizational theory | Expert judgment | Meetings

Tool / Technique	Simple Meaning
Pre-assignment	Some team members are assigned to the project before it officially starts.
Negotiation	The project manager negotiates with functional managers or other parties to obtain required staff.
Acquisition	Hiring or getting people from outside the organization when internal staff are not available.
Virtual Teams	Team members work from different locations using online communication tools.
Multi-criteria Decision Analysis	Selecting team members based on several criteria such as skills, experience, availability, and cost.

Acquire Project Team Outputs The outputs are: Project staff assignment | Resource calendars | Project management plan updates. Developing the project team means improving team members' skills, teamwork, trust, and collaboration so that project performance improves. Tools and Techniques for Developing the Project Team: Interpersonal skills | Training | Team-building activities | Ground rules | Colocation | Recognition and rewards | Personnel assessment tools. Steps to Build a Strong Team: Define the project's goals and objectives clearly and communicate them to the team. | Establish clear roles and responsibilities for each team member. | Foster open communication and encourage collaboration. | Provide training, mentoring, and coaching. | Recognize and reward team members for achievements and contributions. Tuckman Model of Team Development

Stage	Simple Meaning
Forming	Team members meet, understand the project, and learn their roles.
Storming	Conflicts or disagreements may happen as members adjust to working together.
Norming	Team members start accepting rules, roles, and working relationships.
Performing	Team works effectively and focuses strongly on project goals.
Adjourning	Team finishes the project work and separates or moves to other work.

Managing the project team involves tracking team performance, giving feedback, resolving conflicts, motivating team members, and making changes to improve project performance. Tools and Techniques for Managing Project Teams: Observation and conversation | Project performance appraisals | Interpersonal skills | Conflict management. Metrics to Evaluate Team Member Performance

Metric	Simple Meaning
Schedule adherence	How well team members meet project deadlines.
Quality of work	How good the work produced by team members is.
Communication	How well team members communicate with each other and stakeholders.
Budget adherence	How well team members manage project costs.
Stakeholder satisfaction	How well team members meet stakeholder expectations.

Conflict Handling Modes

Conflict Mode	Simple Meaning
Confrontation	Directly face the conflict using a problem-solving approach.
Compromise	Use a give-and-take approach.
Smoothing	Reduce differences and emphasize areas of agreement.
Forcing	Win-lose approach.
Withdrawal	Retreat or withdraw from an actual or potential disagreement.
Collaborating	Combine different viewpoints to build consensus and commitment.

Project Risk Management: Project Risk Management helps the project manager identify possible uncertainties, analyze their impact, and plan responses to reduce negative risks and increase positive opportunities. Negative risk means a possible problem that can harm the project. It may affect project cost, time, quality, scope, or success. Positive risk means an uncertainty that can create a good result. Positive risks are also called opportunities. Risk utility or risk tolerance means the amount of satisfaction or pleasure received from a potential payoff.

Risk Preference	Simple Meaning
Risk-averse	Avoids risk and prefers safer options.
Risk-seeking	Accepts more risk for higher possible reward.
Risk-neutral	Balances risk and payoff.

Project Risk Management Processes: Project Risk Management has six main processes: Plan Risk Management Decide how to approach and plan risk management activities for the project. Identify Risks Determine which risks are likely to affect the project and document their characteristics. Perform Qualitative Risk Analysis Prioritize risks based on their probability and impact. Perform Quantitative Risk Analysis Numerically estimate the effect of risks on project objectives. Plan Risk Responses Decide actions to enhance opportunities and reduce threats to project objectives. Control Risks Monitor identified risks, residual risks, and new risks; carry out response plans and evaluate their effectiveness. Risk Management Plan The main output of planning risk management is the Risk Management Plan. It documents the procedures for managing risk throughout the project. Risk Management Plan Contents: Methodology | Roles and responsibilities | Budget and schedule | Risk categories | Risk probability and impact | Revised stakeholder tolerances | Tracking | Risk documentation. Contingency Plan, Fallback Plan, and Reserves**:** A contingency plan is a predefined action that the project team will take if an identified risk event occurs. Example: If the main server fails, switch to backup server. A fallback plan is used for high-impact risks. It is put into action if the main risk response is not effective. Example: If backup server also fails, move the system to cloud hosting. Contingency reserves are funds or time allowances kept to reduce the risk of cost or schedule overruns. Management reserves are funds kept for unknown risks. Broad Categories of Risk:

Risk Category	Simple Meaning
Market Risk	**Risk related to market demand, competitors, or customers.**
Financial Risk	**Risk related to budget, funding, or financial loss.**
Technology Risk	**Risk related to tools, platforms, systems, or technical complexity.**
People Risk	**Risk related to team members, skills, turnover, or communication.**
Structure / Process Risk	**Risk related to organizational structure, workflow, or poor processes.**

RBS organizes project risks into categories so the team can identify and manage risks more clearly. Identifying risks means understanding what potential events might hurt or enhance the project. Risk Identification Tools and Techniques: Brainstorming | Delphi Technique | Interviewing | SWOT Analysis. Brainstorming A group generates ideas freely without judging them immediately. Delphi Technique Experts provide opinions anonymously, and their answers are summarized until a common view is reached. Interviewing People with similar project experience are interviewed to identify possible risks. SWOT Analysis SWOT identifies Strengths, Weaknesses, Opportunities, and Threats. It helps identify both negative and positive risks. A Risk Register is a document that contains the results of risk management processes. It is often displayed in a table or spreadsheet format and is used to document potential risk events and related information. Qualitative risk analysis means assessing the likelihood and impact of identified risks to determine their priority. Tools and Techniques: Probability / Impact Matrix | Top Ten Risk Item Tracking | Expert Judgment. Quantitative risk analysis numerically estimates the effect of risk on project objectives. It often follows qualitative risk analysis, especially for large and complex projects. Main Techniques: Decision Tree Analysis | Simulation | Sensitivity Analysis. **EMV = Risk Event Probability × Risk Event Monetary Value.** Negative Risk Response Strategies: For negative risks, the project team can use four main strategies: **avoidance, acceptance, transference, and mitigation**. Avoidance means changing the project plan to completely remove the risk. Acceptance means accepting the risk and taking action only if it happens. Transference means shifting the risk impact to another party, such as through insurance or outsourcing. Mitigation means reducing the probability or impact of the risk. Positive Risk Response Strategies**:** For positive risks or opportunities, the project team can use four main strategies: **exploitation, sharing, enhancement, and acceptance**. Exploitation means making sure the opportunity definitely happens. Sharing means working with another party to gain the opportunity. Enhancement means increasing the probability or positive impact of the opportunity. Acceptance means accepting the opportunity if it happens, without actively pursuing it. Residual risks are risks that remain after all response strategies have been implemented. Secondary risks are new risks that happen as a direct result of implementing a risk response. Controlling risks means executing the risk management process, responding to risk events, and keeping risk awareness active throughout the whole project.

Project Stakeholder Management: A stakeholder is any person, group, or organization that is involved in the project or affected by the project. Project Stakeholder Management helps the project manager identify stakeholders, understand their expectations, communicate with them, manage issues, and keep them engaged throughout the project. Project Stakeholder Management Processes: **Identify Stakeholders** Identify everyone involved in the project or affected by it and decide the best ways to manage relationships with them. **Plan Stakeholder Management** Determine strategies to effectively engage stakeholders. **Manage Stakeholder Engagement** Communicate and work with stakeholders to satisfy their needs and expectations, resolve issues, and encourage engagement in project decisions and activities. **Control Stakeholder Engagement** Monitor stakeholder relationships and adjust plans and strategies for stakeholder engagement when needed. Internal stakeholders are people or groups inside the organization.(Project sponsor ,Project team ,Support staff ,Internal customers ,Top management). External stakeholders are people or groups outside the organization who are involved in or affected by the project.(External customers, Competitors ,Suppliers, Government officials , Concerned citizens). A Stakeholder Register is a document that stores important information about project stakeholders. A Power/Interest Grid groups stakeholders based on: Their level of **power/authority**. Their level of **interest/concern** in project outcomes. It has four quadrants. Different quadrants require different management approaches.

Power / Interest Level	Management Strategy	Simple Meaning
High Power / High Interest	**Manage Closely**	These stakeholders are very important. Keep them highly engaged and involved.
High Power / Low Interest	**Keep Satisfied**	They have authority but are not involved daily. Give high-level updates and avoid overloading them.
Low Power / High Interest	**Keep Informed**	They care about the project but have less authority. Give regular updates and collect feedback.
Low Power / Low Interest	**Monitor**	They have low authority and low interest. Monitor them with minimum effort.

A stakeholder management plan explains how stakeholders will be engaged and managed throughout the project. Stakeholder Engagement Levels:

Engagement Level	Simple Meaning
Unaware	Stakeholder does not know about the project or its impacts.
Resistant	Stakeholder knows about the project but resists the change.
Neutral	Stakeholder knows about the project but is neither supportive nor resistant.
Supportive	Stakeholder knows about the project and supports the change.
Leading	Stakeholder knows about the project and is actively engaged in making it successful.

An issue log helps the project team record, monitor, and resolve project issues before they create conflicts or stakeholder dissatisfaction. Issue log may include: Issue ID | Issue description | Owner | Priority | Status | Due date | Resolution. Controlling stakeholder engagement means monitoring stakeholder relationships and adjusting strategies to improve stakeholder involvement and satisfaction.